22 Twenty-five lamps are connected in series to a mains supply. The lamps do not light up when the supply is switched on. A voltmeter gives a zero reading when connected across the third lamp in the set. Which fault cannot be the only one in the circuit?

A The filament of one of the other lamps is broken.

B The filament of the third lamp is broken.

C The fuse in the mains supply has blown.

D There is a break in the wire connecting the set of lamps to the supply.

Ans: **B**

If only B happens, then the pd across the third lamp is the pd across the supply and will not be zero.